

Vetrivel P

+91 90422 13566 — vetrivel13022005@gmail.com — Tamil Nadu, India 636803
github.com/Vetrivel-P2005 — linkedin.com/in/vetrivel-p-144a182b6

About

Computer Science undergraduate at Anna University (MIT Campus) specializing in Web Development and Agentic AI. Strong in problem-solving and collaborative teamwork, with a proven track record of building complex multi-agent frameworks and full-stack applications.

Skills

- **Languages:** C, C++, Java, Python, JavaScript, SQL
- **Web Development:** React.js, Node.js, Express.js, FastAPI, Firebase, REST APIs
- **AI / ML:** Agentic AI, LangChain, LangGraph, LLM Fine-Tuning, Prompt Engineering
- **Databases & Tools:** MongoDB, Git, GitHub, Wireshark, Nmap, ROS 2

Education

Anna University, MIT Campus <i>B.E. in Computer Science and Engineering — CGPA: 8.04 / 10</i>	Chennai, India <i>Expected 2027</i>
Government Model School <i>Higher Secondary Certificate (HSC) — Score: 565 / 600 (94.17%)</i>	Dharmapuri, India <i>2023</i>
Government High School <i>Secondary School Leaving Certificate (SSLC)</i>	Dharmapuri, India <i>2021</i>

Projects

Hierarchical Agentic Framework for ICU Respiratory Care • Developed an AI-powered clinical decision support system for ICU respiratory care using Agentic AI and biomedical LLMs. • Built a multi-agent workflow with LangGraph for patient monitoring, clinical reasoning, and automated treatment recommendations. • Fine-tuned BioMistral-7B using QLoRA on ICU patient datasets and integrated time-series analysis for vital sign tracking. • Developed a FastAPI backend paired with a React.js dashboard for real-time recommendation visualization. • Tech Stack: Python, LangGraph, LangChain, BioMistral-7B, QLoRA, FastAPI, React.js.	<i>Agentic AI Research</i>
ML-Based Network Packet Classification • Built machine learning models to classify network packets (TCP, UDP, ICMP) from live Wireshark PCAP datasets. • Handled feature extraction, data preprocessing, and model evaluation utilizing Scikit-learn. • Tech Stack: Python, Scikit-learn, Wireshark.	<i>Networking & Machine Learning</i>
Network Vulnerability Scanner • Developed a modular Python-based script using Nmap to automate network reconnaissance and active port scanning. • Streamlined security assessment workflows to detect open ports and flag basic network vulnerabilities. • Tech Stack: Python, Nmap.	<i>Cybersecurity</i>

Certifications

Web Development Internship Certification	<i>Glowlogics Solutions (2025)</i>
Cybersecurity and Ethical Hacking Internship Certification	<i>Glowlogics Solutions (2025)</i>

Languages

- Tamil (Native), English (Professional)